ThoughtFlow-FP8: A Mixed Gate for Phase-Aware KV Retention

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Abstract

ThoughtFlow-FP8 tests whether explicit anchor/phase retention plus byte-aware FP8-style budgeting can preserve useful reasoning context better than matched-budget KV-retention proxies. Synthetic traces support the intuition that phase markers and anchors can be protected, but real retained-context quality weakens the branch. On 24 distilgpt2 traces at a 0.20 keep fraction, a bounded successor policy, ThoughtFlow-saliency-recent, improves over the original phase-only policy and beats LongFlow-like and ThinKV-like proxies, but narrowly loses to an R-KV-like retained-prefix proxy on continuation NLL (3.434 vs 3.419). We therefore present this as a mixed workshop gate, not a positive method. The next useful step is a sharper hidden/KV saliency policy; GPU sparse-KV work is not justified until that Mac-local proxy clears.

1 Policy Family

The original policy protects anchors and phase labels. The first successor adds a small recent-token reserve. The second successor protects anchors, reserves half the retained budget for recent tokens, and lets phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. This tests whether the failure was caused by over-preserving interpretable markers that do not matter for immediate continuation likelihood.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1

Table 1: Retained-context continuation proxy on 24 distilgpt2 traces. Lower NLL is better. Full context is not a matched-budget baseline.

2 Decision

The branch is mixed but not revived as a positive method. ThoughtFlow-saliency-recent repairs most of the phase-only failure and nearly ties the strongest matched-budget proxy, but it does not beat R-KV-like. A paper-ready claim would require a hidden/KV saliency policy that beats R-KV-like, LongFlow-like, and ThinKV-like rows at matched bytes, then validates under real sparse-KV or cache-dropping execution.

3 Artifacts

- `experimental/thoughtflow_fp8/phase2/perplexity_impact_proxy.md`

- experimental/thoughtflow_fp8/phase2/hidden_saliency_retention_probe.md
- experimental/thoughtflow_fp8/phase2/real_trace_retention_sweep.md
- experimental/thoughtflow_fp8/paper/colm_workshop_shell.md